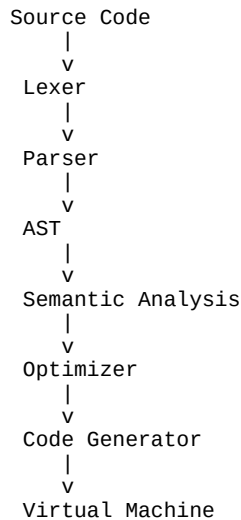


MyOwnCompiler - Day 1 Notes

Compiler Engineering Journal

Day 1 established the foundation of the compiler project: architecture, build system, token infrastructure, and workflow.

Compiler Pipeline



Repository Structure

```
MyOwnCompiler
├── docs
├── tests
├── src
│   ├── lexer
│   ├── parser
│   ├── ast
│   ├── semantic
│   ├── optimizer
│   ├── codegen
│   ├── vm
│   └── main.cpp
├── README.md
├── .gitignore
└── CMakeLists.txt
```

What is a Token?

The lexer converts raw characters into meaningful units called tokens.

```
let x = 10;
```

becomes

```
LET
IDENTIFIER(x)
EQUAL
NUMBER(10)
SEMICOLON
EOF
```

TokenType

```
enum class TokenType
{
    Identifier,
    Number,
    Plus, Minus, Star, Slash,
```

```
        Equal,  
        Semicolon,  
        Let, If, While, Print,  
        EndOfFile,  
        Unknown  
    };
```

Token Structure

```
struct Token  
{  
    TokenType type;  
    std::string lexeme;  
};
```

Example Token

```
Token {  
    type = Identifier,  
    lexeme = "age"  
}
```

Concepts Learned

Compiler pipeline, tokenization, enum class, constructors, std::move, CMake, project architecture, Git workflow.

Day 1 Result

Commit: 205915a - Phase 0 Day 1: project setup and token infrastructure

Day 2 Preview

Implement a real lexer that converts source code into a vector of tokens.